

Week 4 — Superposition Component Library

Geophysics 3A: Potential Fields & Geothermics

March 10, 2026

How to use

Select 2–3 components to build a plausible local cross-section. For each, choose a realistic $\Delta\rho$ (from Density Worksheet), depth, and width. Overlay normalized response shapes, scale amplitudes, shift laterally, and *sum* to obtain a composite gravity profile.

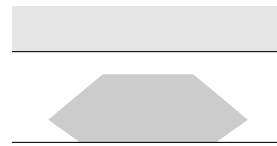
A) Basement Step with Sedimentary Wedge

- **Geology:** Sediments thickening toward a normal fault; basement uplift on one side.
- **Kernel:** Edge-dominated (polygon/step); asymmetry likely.
- **Params:** $\Delta\rho$ (sediment vs. basement): $\sim 200\text{--}600\text{ kg m}^{-3}$; depth to top: 0.5–2 km; wedge width: 1–5 km.



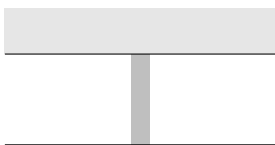
C) Felsic Pluton (Broad Polygon)

- **Geology:** Large, less dense body emplaced at depth.
- **Kernel:** Broad, low-amplitude positive or negative depending on host; gentle gradients.
- **Params:** $\Delta\rho$: -100 to $+200\text{ kg m}^{-3}$; width: 2–10 km; depth to top: 1–5 km.



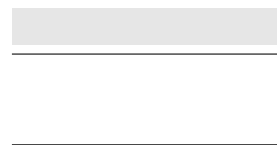
B) Mafic Dike (2D Cylinder Proxy)

- **Geology:** Narrow, dense intrusive cutting felsic country rock.
- **Kernel:** Narrow positive peak; half-width gives depth cue.
- **Params:** $\Delta\rho$ (mafic–felsic): $\sim 200\text{--}400\text{ kg m}^{-3}$; width: 50–300 m; depth to top: 0.1–1 km.



D) Regolith/Weathered Cap (Thin Negative Slab)

- **Geology:** Near-surface low-density layer.
- **Kernel:** Thin slab; small, relatively localized negative anomaly.
- **Params:** $\Delta\rho$: -100 to -400 kg m^{-3} ; thickness: 5–50 m; lateral extent: 0.5–2 km.



Notes

Use the *Formula Sheet* for scale/shape cues (half-width, edge effects) and superposition rules. Combine components judiciously to avoid overfitting.